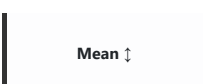
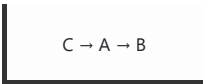
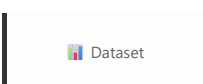


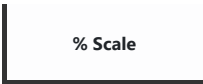
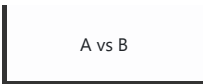
# Seaborn Parameter Impact & Dummy Visual Reference

## 1. sns.barplot()

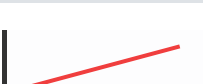
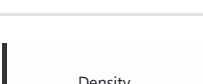
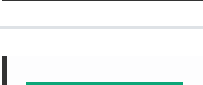
| PARAMETER | MEANING              | EXAMPLE                             | VISUAL IMPACT / OUTCOME                                     | DUMMY VISUAL                                                                          |
|-----------|----------------------|-------------------------------------|-------------------------------------------------------------|---------------------------------------------------------------------------------------|
| data      | DataFrame source     | <code>data=df</code>                | Links function directly to active dataset columns.          |    |
| x, y      | Axis variables       | <code>x='Airline', y='Price'</code> | Maps categories on X-axis and metric values on Y-axis.      |    |
| hue       | Grouping variable    | <code>hue='Class'</code>            | Splits main category into sub-bars mapped by color.         |    |
| palette   | Color map            | <code>palette='viridis'</code>      | Applies a custom thematic gradient color set.               |    |
| errorbar  | Confidence interval  | <code>errorbar=('ci', 95)</code>    | Draws vertical error indicator lines representing variance. |  |
| estimator | Statistical function | <code>estimator=np.mean</code>      | Changes aggregation mode (e.g., switches from mean to sum). |  |
| order     | Category order       | <code>order=['A', 'B', 'C']</code>  | Reorders the category sequence along the categorical axis.  |  |
| capsize   | Error bar caps       | <code>capsize=0.2</code>            | Adds terminal horizontal caps at the tips of error lines.   |  |

## 2. sns.countplot()

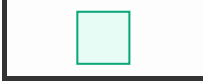
| PARAMETER | MEANING              | EXAMPLE                  | VISUAL IMPACT / OUTCOME                                      | DUMMY VISUAL                                                                          |
|-----------|----------------------|--------------------------|--------------------------------------------------------------|---------------------------------------------------------------------------------------|
| x / y     | Categorical variable | <code>x='Airline'</code> | Counts occurrences of items automatically within categories. |  |
| data      | DataFrame source     | <code>data=df</code>     | Binds the count routine directly to the input dataset.       |  |

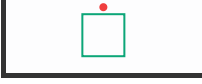
| PARAMETER | MEANING             | EXAMPLE                       | VISUAL IMPACT / OUTCOME                                       | DUMMY VISUAL                                                                        |
|-----------|---------------------|-------------------------------|---------------------------------------------------------------|-------------------------------------------------------------------------------------|
| hue       | Grouping variable   | <code>hue='Class'</code>      | Breaks down each count bar into sub-group segments.           |  |
| palette   | Color map           | <code>palette='Set2'</code>   | Applies distinct categorical color styles.                    |  |
| stat      | Statistic aggregate | <code>stat='percent'</code>   | Switches display values from raw counts to percentage ratios. |  |
| order     | Category order      | <code>order=['A', 'B']</code> | Sorts and sequences category bins manually.                   |  |

### 3. sns.histplot()

| PARAMETER | MEANING            | EXAMPLE                       | VISUAL IMPACT / OUTCOME                                     | DUMMY VISUAL                                                                          |
|-----------|--------------------|-------------------------------|-------------------------------------------------------------|---------------------------------------------------------------------------------------|
| x / y     | Data variable      | <code>x='Price'</code>        | Computes distribution ranges along the chosen axis.         |   |
| bins      | Number of bins     | <code>bins=30</code>          | Controls resolution and granularity of histogram bars.      |  |
| kde       | Density curve      | <code>kde=True</code>         | Overlays a smooth continuous density line estimation curve. |  |
| stat      | Aggregation metric | <code>stat='density'</code>   | Normalizes bar heights into density or probability metrics. |  |
| hue       | Grouping variable  | <code>hue='Airline'</code>    | Overlays distribution curves or bars for separate groups.   |  |
| multiple  | Layout for hue     | <code>multiple='stack'</code> | Stacks grouped distribution bars on top of each other.      |  |

### 4. sns.boxplot()

| PARAMETER | MEANING                 | EXAMPLE                             | VISUAL IMPACT / OUTCOME                          | DUMMY VISUAL                                                                          |
|-----------|-------------------------|-------------------------------------|--------------------------------------------------|---------------------------------------------------------------------------------------|
| x, y      | Categorical & Numerical | <code>x='Airline', y='Price'</code> | Draws quartile distribution boxes with whiskers. |  |

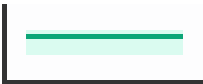
| PARAMETER | MEANING            | EXAMPLE                     | VISUAL IMPACT / OUTCOME                                   | DUMMY VISUAL                                                                        |
|-----------|--------------------|-----------------------------|-----------------------------------------------------------|-------------------------------------------------------------------------------------|
| hue       | Grouping variable  | <code>hue='Class'</code>    | Puts side-by-side comparative box plots per category.     |  |
| notch     | Notched confidence | <code>notch=True</code>     | Adds confidence interval notches around median lines.     |  |
| fliersize | Outlier size       | <code>fliersize=3</code>    | Modifies marker dimensions for plotted outlier points.    |  |
| showmeans | Show mean          | <code>showmeans=True</code> | Displays an arithmetic mean marker symbol inside the box. |  |

## 5. sns.scatterplot()

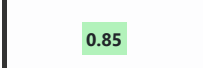
| PARAMETER | MEANING               | EXAMPLE                              | VISUAL IMPACT / OUTCOME                                     | DUMMY VISUAL                                                                          |
|-----------|-----------------------|--------------------------------------|-------------------------------------------------------------|---------------------------------------------------------------------------------------|
| x, y      | Numeric coordinates   | <code>x='Duration', y='Price'</code> | Plots coordinate intersection dots across 2D axes.          |  |
| hue       | Color grouping        | <code>hue='Airline'</code>           | Assigns distinct colors to points based on group labels.    |  |
| size      | Marker size grouping  | <code>size='Stops'</code>            | Scales individual point radiuses dynamically based on data. |  |
| style     | Marker shape grouping | <code>style='Class'</code>           | Changes point geometries (circles, crosses, squares).       |  |
| alpha     | Transparency          | <code>alpha=0.7</code>               | Adjusts point opacity to reveal dense overlapping clusters. |  |

## 6. sns.lineplot()

| PARAMETER | MEANING             | EXAMPLE                          | VISUAL IMPACT / OUTCOME                              | DUMMY VISUAL                                                                          |
|-----------|---------------------|----------------------------------|------------------------------------------------------|---------------------------------------------------------------------------------------|
| x, y      | Numeric coordinates | <code>x='Date', y='Price'</code> | Connects chronological points with continuous paths. |  |

| PARAMETER | MEANING             | EXAMPLE                    | VISUAL IMPACT / OUTCOME                                      | DUMMY VISUAL                                                                        |
|-----------|---------------------|----------------------------|--------------------------------------------------------------|-------------------------------------------------------------------------------------|
| hue       | Color grouping      | <code>hue='Airline'</code> | Draws multiple parallel trend lines distinguished by color.  |  |
| markers   | Show data points    | <code>markers=True</code>  | Places visible point nodes along the trajectory line.        |  |
| errorbar  | Confidence interval | <code>errorbar=None</code> | Shades confidence interval bands around the line trajectory. |  |

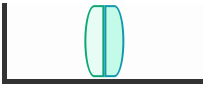
## 7. sns.heatmap()

| PARAMETER  | MEANING           | EXAMPLE                       | VISUAL IMPACT / OUTCOME                                      | DUMMY VISUAL                                                                          |
|------------|-------------------|-------------------------------|--------------------------------------------------------------|---------------------------------------------------------------------------------------|
| data       | Matrix dataset    | <code>data=corr_matrix</code> | Renders 2D matrix grids color-coded by value intensity.      |    |
| annot      | Write values      | <code>annot=True</code>       | Prints actual numeric values directly inside each grid cell. |   |
| cmap       | Color map theme   | <code>cmap='coolwarm'</code>  | Changes the underlying color gradient spectrum profile.      |  |
| linewidths | Cell grid borders | <code>linewidths=0.5</code>   | Adds white or colored grid border separation lines.          |  |

## 8. sns.pairplot()

| PARAMETER | MEANING            | EXAMPLE                    | VISUAL IMPACT / OUTCOME                                    | DUMMY VISUAL                                                                          |
|-----------|--------------------|----------------------------|------------------------------------------------------------|---------------------------------------------------------------------------------------|
| data      | DataFrame source   | <code>data=df</code>       | Generates an expansive multi-plot matrix grid.             |  |
| hue       | Color grouping     | <code>hue='Airline'</code> | Color-codes all subplots across the grid by categories.    |  |
| kind      | Non-diag plot type | <code>kind='reg'</code>    | Switches subplot types (e.g., from scatter to regression). |  |

## 9. sns.violinplot()

| PARAMETER | MEANING                 | EXAMPLE                             | VISUAL IMPACT / OUTCOME                                      | DUMMY VISUAL                                                                        |
|-----------|-------------------------|-------------------------------------|--------------------------------------------------------------|-------------------------------------------------------------------------------------|
| x, y      | Categorical & Numerical | <code>x='Airline', y='Price'</code> | Draws symmetrical kernel density distribution shapes.        |  |
| split     | Split violins for hue   | <code>split=True</code>             | Cuts violins in half to compare two sub-categories directly. |  |
| inner     | Internal representation | <code>inner='quartile'</code>       | Adds miniature box plot profiles inside the violin core.     |  |

## 10. sns.regplot()

| PARAMETER | MEANING                  | EXAMPLE                              | VISUAL IMPACT / OUTCOME                                        | DUMMY VISUAL                                                                          |
|-----------|--------------------------|--------------------------------------|----------------------------------------------------------------|---------------------------------------------------------------------------------------|
| x, y      | Numeric input data       | <code>x='Duration', y='Price'</code> | Plots scatter coordinate points with a fitted model trendline. |    |
| ci        | Confidence interval size | <code>ci=95</code>                   | Shades regression uncertainty bands around the trend line.     |  |
| order     | Polynomial regression    | <code>order=2</code>                 | Curves the fitted model line to capture non-linear trends.     |  |